

Question 1:

Two distinct numbers a and b are selected at random from $1, 2, 3, \dots, 50$. The probability that their product ab is divisible by 3 is

- (A) $8/25$
- (B) $561/1225$
- (C) $664/1225$
- (D) $272/1225$

Question 2:

If a random variable x has the probability distribution

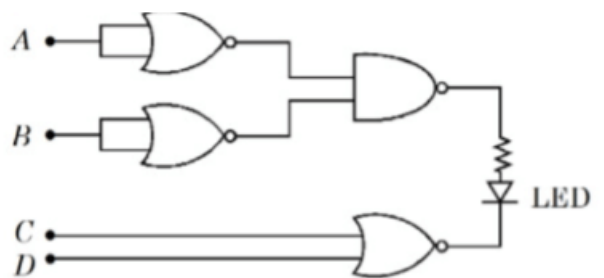
x	0	1	2	3	4	5	6	7
$P(x)$	0	$2k$	k	$3k$	$2k^2$	$2k$	$k^2 + k$	$7k^2$

then $P(3 < x \leq 6)$ is equal to

- (A) 0.22
- (B) 0.33
- (C) 0.34
- (D) 0.64

Question 3:

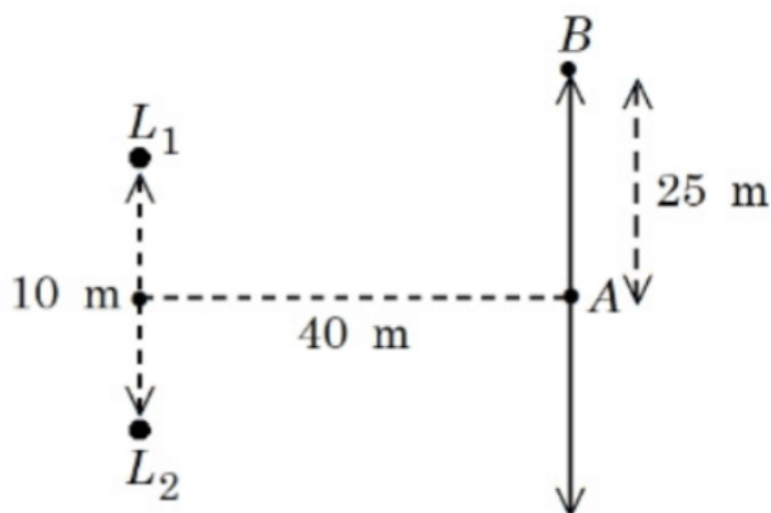
Find the correct combination of A, B, C and D inputs which can cause the LED to glow.



- (A) 0100
- (B) 1000
- (C) 0011
- (D) 1101

• Question 4:

Two loudspeakers (L_1 and L_2) are placed with a separation of 10m, as shown in the figure. Both speakers are fed with an audio input signal of the same frequency with constant volume. A voice recorder, initially at point A, at equidistance to both loudspeakers, is moved by 25m along the line AB while monitoring the audio signal. The measured signal was found to undergo 10 cycles of minima and maxima during the movement. The frequency of the input signal is _____ Hz.

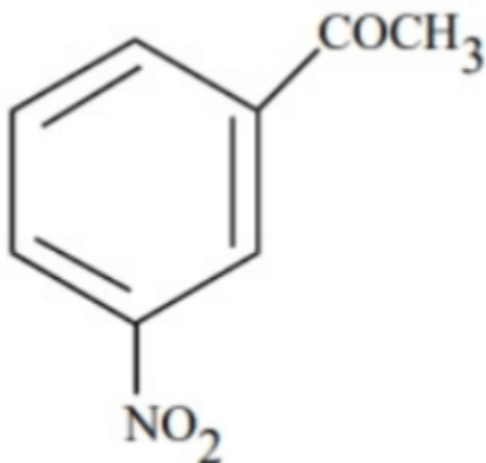


(Speed of sound in air is 324m/s and $\sqrt{5}=2.23$)

Question 5:

Given below are two statements:

Statement I: Benzene is nitrated to give nitrobenzene, which on further treatment with $\text{CH}_3\text{COCl}/\text{AlCl}_3$ will give the product shown.



Statement II: $-\text{NO}_2$ group is a meta-directing and deactivating group.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Statement I is correct but Statement II is incorrect
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is incorrect but Statement II is correct
- (D) Both Statement I and Statement II are correct

Question 6:

Match the LIST-I with LIST-II.

List-I		List-II	
Thermodynamic Process		Magnitude in kJ	
A.	Work done in reversible, isothermal expansion of 2 mol of ideal gas from 2 dm^3 to 20 dm^3 at 300 K.	I.	4
B.	Work done in irreversible isothermal expansion of 1 mol ideal gas from 1 m^3 to 3 m^3 at 300 K against a constant pressure of 3kPa.	II.	11.5
C.	Change in internal energy for adiabatic expansion of a 1 mol ideal gas with change of temperature = 320 K and $\bar{C}_V = \frac{3}{2} R$.	III.	6
D.	Change in enthalpy at constant pressure of 1 mol ideal gas with change of temperature = 337 K and $\bar{C}_p = \frac{5}{2} R$.	IV.	7

Choose the correct answer from the options given below:

- (A) A-III, B-II, C-IV, D-I
- (B) A-II, B-I, C-III, D-IV
- (C) A-I, B-II, C-III, D-IV
- (D) A-II, B-III, C-I, D-IV